

EXP NO: 10

STUDY OF CLOCKS RS, D,
JK FLIP FLOP

DATE:

15/11/22

AIM:

To study the truth table RS, D, JK flip flop.

COMPONENTS REQUIRED:

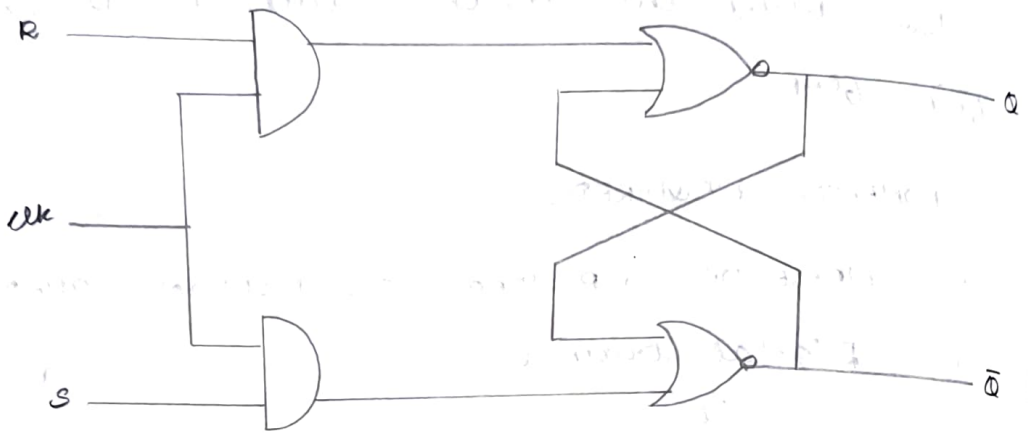
S.NO	NAME OF COMPONENT	SPECIFICATION	QUANTITY
1	Digital trainer kit	-	1
2	AND gate	IC 7408	1
3	NOT gate	IC 7404	1
4	NAND gate	IC 7400	1
5	NOR gate	IC 7486	1
6	Connecting wires	-	As required

THEORY:

Flip flop and latches are used data storage element. A flip flop stores single bit single digit of data that represent '2ab'. Such data storage can be used for storage of gate and such a circuit is described as ~~sequential~~ logic when used in finite state machine the output and next state depend not only it current output also state and hence previous input timed input signal to some reference timing signals.



LOGIC DIAGRAM:



TRUTH TABLE

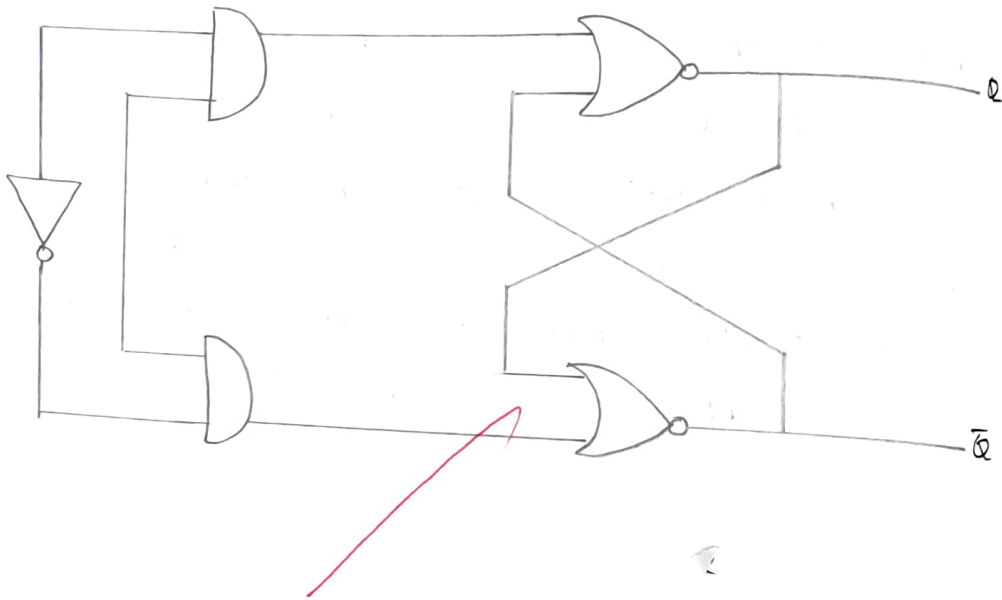
Q	R	S	$\bar{Q}$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	In determinate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	In determinate

RS flip flop:

This form of flip flop input one or more output lines that always present is labeled as a generally second output called 'a'. Will also represent one output here $S=1, S=0$ used to reset the device to $Q=0$, state this condition normally is in actual circuit operation.



LOGIC DIAGRAM:



TRUTH TABLE:

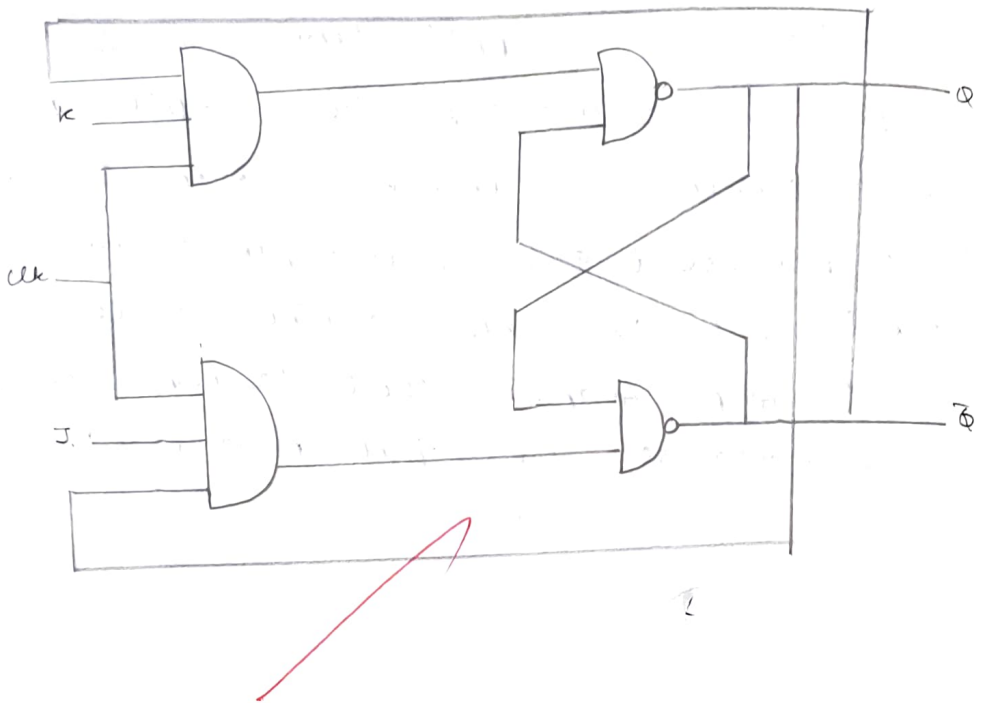
D	Q	Q̄
0	0	1
1	1	0

D flip flop:

On same application the S and K input will always be complementary $K=0$ taken $S=1$ and $K=1$ when $S=0$. That can processed $S=1$ because pin connection can minimized. The chip circuit device known as D flip flop called transparent flip flop because the input is $Q=1$ when $D=1$, $Q=0$ when $D=0$.



LOGIC DIAGRAM:



TRUTH TABLE:

Q	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

JK flip flop:

This table circuit has two going input along with a clock input the voltage level of gates determines output state to which to clock output will shift the flip flop order $J=1, K=0$ and the reset condition when $S=0, R=1$ when both $J=K=0$ then condition is called both $J=K=1$. The clock input cause the flip flop to toggle. Complement of previous state when both $J=K=1$. and the clock pulse is calculated for a long time taken the output flip flop keep toggling which is race condition this can be choosing the clock pulse entered. than the propagation delay of flip flop.

PROCEDURE:

* The required IC are placed on bread board of digital transistor.

* Connections are given as per the circuit diagram.

* IC must be bared with V_{CC} and ground connector are into their respected pins.



*) The input are get are taken from the input terminals of the trainer kit.

*) The output gives to the LED assigned terminals.

*) The corresponding logic operation for all the gates are connected and checked per the truth table.

RESULT:

Thus the truth table such a JK flip flop, RS flip flop and D flip flop is studied and verified successfully.

